Toward Adaptive, Explainable Multimodal Triage in the Emergency Department: A Review and Proposed Architecture

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Abstract*— Manual observation and a limited number of parameterized severity scoring systems are still very much the main modalities used in the emergency department (ED) triage, while information lag is slow, subjective results and can be difficult to adjust to new clinical evidence. Many distinct aspects of this challenge have been solved by the use of artificial intelligence (AI): Individual solutions for electrocardiogram (ECG) analysis, chest radiograph analysis, risk modelling based on structured vital signs (VTS) and large-language-model (LLM) based triage have been reported throughout the literature. The aim of this paper is to offer a structured review of 59 recent studies from five thematic groups namely: (A) AI-assisted ED triage, (B) multimodal medical AI, (C) fusion of imaging and clinical data, (D) missing-modality robustness, (E) explainable AI (XAI) and clinician trust (2023–2026). There are three gaps that are common: Integration of multi-modality data is inadequate, risk estimation is static and non-adaptive, and incorporation of the incomplete data is inadequate, with low degree of interpretability for the clinician. We combine these results into a proposed multimodal adaptive, explainable system for triage, which fuses symptoms, vital signs, ECG and chest X-ray data for an explainable modality-aware fusion module, inference-dynamically updates the triage risk score, and provides a result that is easily read by clinicians. The review brings up two salient points that have received scant attention to date in the development of clinically viable AI-powered triage systems: absence of modalities and fairness.Keywords— multimodal AI, emergency triage, explainable AI, missing modality, clinical decision support, risk prediction

I. INTRODUCTION

Patient prioritization in the ED is a key component of clinical outcomes that is distinct and a necessary part of the process to be efficient and effective. Conventional triage relies on rule-based severity scoring and manual clinical judgement, but these are sub-optimal when applied to very large numbers of patients and are difficult to modify to accept heterogeneous and constantly changing information such as laboratory values, electrocardiograms and X-rays [3, 11]. While significant advances have been made over the last three years in AI research for individual sub-problems in triage, such as tasking ECG for risk prediction [1], chest X-ray use for interpretation [36]-[38] and the incorporation of language inputs using LLM supported conversational triage systems [17], [18], a comprehensive realization of a system that jointly fused these modalities has yet to be achieved. There has been an extensive literature review on the basis of this type of system. To narrow the scope of the review to the four technical challenges of an AI diagnostic based on multimodal approach.

triage system being developed, we collectively group the challenges into four specific goals (O1) multimodal risk/triage assessment; (O2) dynamic and adaptive triage; (O3) explainable and confidence-aware prediction; (O4) robustness to missing modalities, with a cross-cutting query of fairness-aware evaluation (F). The contributions of this review are: (i) taxonomy of 59 relevant literature studies from a multimodal triage perspective based on 5 clusters; (ii) explicit mapping of literature gaps and the 4 design objectives; and (iii) mapping the gaps in each cluster and identifying the design objectives within each. Proposed system architecture based on reviewed. evidence, in relation with adaptive multimodal triage with explainability and missing-modality tolerance.

II. LITERATURE REVIEW

The distribution of the 59 reviewed studies is summarized in Fig. 1 which is divided into five thematic areas, discussed in Sections II-A–II-E.

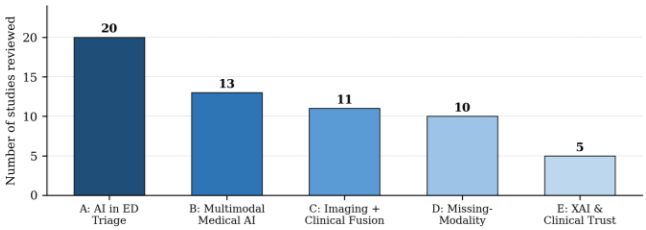


Fig. 1. Data display for reviewed studies in the thematic clusters A – E.

A. AI-Assisted Emergency Department Triage

There is an increasing and large body of work using machine learning (ML) and deep learning in ED Triage. The authors of Elhussain et al. [1] performed a systematic review of 108 studies on ECG-AI, facing repeated problems with internal and external validation, calibration, robustness, and even generalizability, and therefore not recommending the use of ECG as a "stand-alone" diagnosis, but as "one of several signals. On the contrary, AI structured triage has been recognized as not being properly applied in clinical work, its accuracy in the dispositions, not to mention in replacing structured clinical assessment [2,3,18,19]. Dedicated algorithms can be used to build deep learning classifiers which provide better classification accuracy than general purpose LLMs, but can only be interested in a single classification label, rather than a dynamically varying classification score from multiple sources.

There are a number of studies that focus on particular populations or symptom groups such as paediatric triage for admission, cardiac-symptom triage with only structured symptoms [5], [9], [10]. Nursing- and safety-oriented reviews [6]-[8] are all on the same page: that AI has a potential to optimally enhance triage consistency and yet clinical integration, transparency and benchmarking in terms of safety and control against experts' judgment are limited. More specifically, the reviewed work on ED triage [11] in Porto highlights the lack of utilization of explainable AI (XAI) and feature engineering, and even mentions the potential of high bias, risks in many models explored, thereby reinforcing the need for a fusion approach combined with XAI and the evaluation of fairness found here. The concept of answering questions on completion of the record or alternatively optimizing for dispositions when the patient enters the hospital and continually updating the record with new records and not new cases for urgency remains, as does the lack of real-time, prospective, multi-centre prediction of dispositions [14], [12], [13], [15],[16]. Finally, a 2025 review of AI for multimodal ED triage [20] explicitly identifies data dependency, ethical issues, and inconsistent performance as controversial issues that have yet to be addressed.

B. Multimodal Medical AI

The second cluster pertains to multimodal AI in medicine in more general terms. The foundation-model approach [21] is an approach that can be used to achieve zero-shot multimodal, multilingual clinical diagnosis but has the limitation of narrow focus on the diagnostic reporting. The survey papers all highlight a common set of open problems: heterogeneous data fusion is a challenging technical problem and – most importantly for this review – missing modalities are typically not considered, leading to the neglect of incomplete cases in the background and the introduction of selection bias, thus hampering generalizability [23]-[25]. This extends into more safety, fairness and responsible-adoption issues in the light of policy/ethics papers [26] and technical issues of data fragmentation, synchronisation and interpretation in the light of broader landscape papers [27] as key issues that remain.

The research efforts that develop multimodal fusion for certain clinical tasks also involve the use of multimodal fusion in the ophthalmic diagnosis scene with agentic interaction [30] and multimodal clinical decision workflow based LLM for triage and referrer. Predicting EHR [32] and [33] using a mixture-of-experts agentic architecture [31] has been proposed. These systems all share a general trait of being diagnosis or reporting oriented – few explicitly translate multimodal fused evidence into an urgent/priority signal which is continuously updated; clinical-workflow papers [29] have pointed out that the motto of AI systems should be to better emulate the sequential, dynamic nature of real clinical encounters, a step that directly leads to an adaptive risk-update design.

C. Fusion of Imaging and Clinical Data

A very similar group of experts focus on leveraging only structured clinical information in conjunction with medical Imaging (mainly chest radiography). The proposed system is similar to Wu et al. [34] which uses chest X-ray and EHR to screen patients with acute heart-failure based on disease rather than a general ED urgency. The concept of using combination of imaging plus clinical parameters, also exists in COPD staging [35] and pneumonia diagnosis using multimodal reference data [37]. Several studies only consider the quality of interpretation of CXRs [36], and the help provided by radiologists [39], [38] or reduction of radiologist workload [38] but don't include this signal in a larger prioritization stream. Recently, methods based on foundation models and diffusion imaging have been developed to improve the quality of the representations but they still operate framewise, without making use of other patient information such as a chest X-ray (CXR). Summary papers [42] and human-versus-AI studies [43] once again emphasize that generalization and the ability to integrate into real clinical pipelines (but not simply a growth in image-classification accuracy) continues to be the biggest hurdle. Also, the fact that data from a single institution, without external validation, are always used was pointed out in a multimodal osteoporosis-screening study [44]. This simply highlights the importance of the need to have evaluation protocols that are cross validated and tested externally.

D. Missing-Modality Robustness

The cluster that is most relevant to Objective O4 and least investigated and acted upon gap in the literature is this cluster. Wang et al. [45] explore the missing-modality enabled fusion for X-ray and radiology reports and structured data, whereas Liaqat et al. [46] develop a framework explicitly considering the case of missing modalities. What both Ruffini et al. [47] and other reviews of multimodal missing data in healthcare [48] have emphasized is that modalities might not be missing in all patients, but are instead the exception. Many papers in the image domain repeat this claim with slight variations [49]: Incomplete multimodal medical images are often not included in the downstream task (e.g., discarding the affected patient) or they cause significant impairment of the downstream task, without the need of training a separate model for each missing-modality configuration [50]. Dedicated architectural strategies such as the meta-completion network [51] and the shared-specific feature modelling [52] and modality-masked diffusion synthesis [53] or disentangled shared/modality-specific representations [54] are required to process these incomplete images. This group allows for a design choice of modality-aware fusion - that is, one or more indicators are made explicit, and the risk estimation the proposed system can produce is not simply ignored when there is no ECG or chest X-ray data, but rather it is actually used.

E. Explainable AI and Clinical Trust

The final group is on interpretability, which is an Objective 3 element. The authors of Pahud de Mortanges et al. [55] suggested that inputs to XAI model and the risk estimate may change over time and, in this case, XAI methods should be adapted, hierarchical and uncertainty aware. However, good explanations do not necessarily lead to proper clinical trust and reliance [56]. One comparative study [57] demonstrates that explanations written in natural language (as opposed to raw SHAP explanations) are more easily accepted by clinicians, and a scoping review of clinicians' perceptions of XAI in the context

of hospital decision-support systems [58] concludes that the technical visualizations generated by SHAP can be confusing and/or time-consuming, and that clinicians prefer simple, visual, clinically relevant explanations. A more thorough investigation of the subject of XAI in the medical field reveals that there is a lingering disconnect between existing explainability approaches and their application in actual clinical settings [59]. The results encourage the use of either SHAP or Grad-CAM with the label in the augmentation plane and making only clinician-readable factors visible in the final dashboard (e.g., SpO_2 84%, HR 145 bpm, opacity of the right-lower-lobe).

III. SYNTHESIS OF RESEARCH GAPS

TABLE I. MAPPING OF LITERATURE-IDENTIFIED GAPS TO DESIGN OBJECTIVES

Obj.	Identified gap in prior work	Rep. refs.
O1	Single-modality models dominate; multimodal fusion is rare or diagnosis-only, not urgency-oriented	[1], [3], [21], [34]
O2	Triage/admission prediction is largely retrospective and static; risk is not re-estimated as new data arrives	[13], [15], [16], [29]
O3	Explanations are technically valid but not clinically legible; low trust without simplified, factor-based output	[55], [57], [58]
O4	Incomplete modalities are common but usually handled by discarding patients rather than adapting the model	[23], [45], [48], [49]
F	Bias and fairness evaluation is inconsistently reported despite ethical concern	[11], [20], [26], [28]

Table I shows the relationship of the gaps in the literature identified above with the four design objectives (O1 – O4) and the fairness concern that cuts across all the four design objectives (F, cross-cutting concern).

There are three cross cutting observations. First, most AI that is used for ED triage (Section II-A) is single-modality or, in the multimodality case, disease-specific, rather than a general urgency one (Section II-C). Second, neighboring imaging applications, including MRI synthesis and tumor segmentation, are technically ripe for missing-modality handling, but have not been widely used in ED-triage applications. Third, research on the topic of explainability (Section II-E) has focused not only on how to interpret the raw feature-attribution, but to be more helpful to their clinicians when communicating. This change is away from raw SHAP value reporting toward simpler reporting on contributing factors, bringing known methods of explainability more in line with current evidence in clinical use.

IV. PROPOSED ARCHITECTURE

Based on the detected issues, the multimodal triage architecture is proposed in Fig. 2. Classical ML (Random Forest/ XG-Boost) and NLP/LLM components are used for tackling the fusion gap identified in Section II-B, with ECG and chest X-ray signals each processed by a dedicated model, a 1-D CNN and a vision CNN respectively, as was found in imaging-specific evidence identified in Section II-C. To satisfy Objective O4 (in the absence of an ECG or X-ray, the fusion layer generates a risk estimate from the remaining modalities), it is applied specifically to agree with the design concept of studies with missing modalities [45], [46], [51], [52]. The fused representation is then passed into an adaptive risk-scoring module for calculating priority in the event of new observations (Objective O2), and an XAI layer (either internally through SHAP or through factors that are explicitly-readable by the clinician, or externally) that satisfies Objective O3 based on evidence (Section II-E). The Clinician Dashboard presentation also provides a priority (Red/Orange/Yellow/Green) color indicator and confidence, plus top contributing factors.

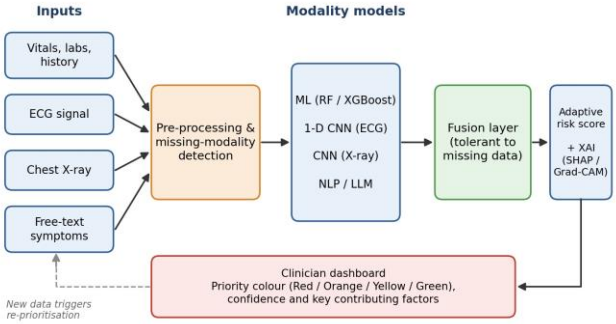


Fig. 2. Proposed adaptive, explainable, missing-modality-tolerant multimodal triage architecture.

V. ADAPTIVE MULTIMODAL TRIAGE WORKFLOW

The architecture of Fig. 2 is broken down into an operational, 7-step pipeline in Fig.3. In the modality-specific processing (Steps 3), classical ML is used for handling blood pressure and other vital/labs, NLP/LLM for free-text symptoms, and a vision model for imaging, as evidenced in respective sections II-A–II-C. A patient is then subjected to multimodal fusion (Steps 4) and risk-score generation (Steps 5), accessible even in the absence of a modality, consistent with the modality-specific evidence in respective sections; critically, re-enters Steps 3 whenever new observations are produced (e.g., a delayed lab result or an X-ray taken after initial assessment) — operationalizing Objective O2 and directly addressing the static-triage limitation raised in [13], [15], [16], [29]. Step 7 adds a clinician-readable explanation to each and every change of priority, enabling a clear understanding of reprioritization as opposed to a black box one, based on human-factors evidence provided by Section II-E [56-58].

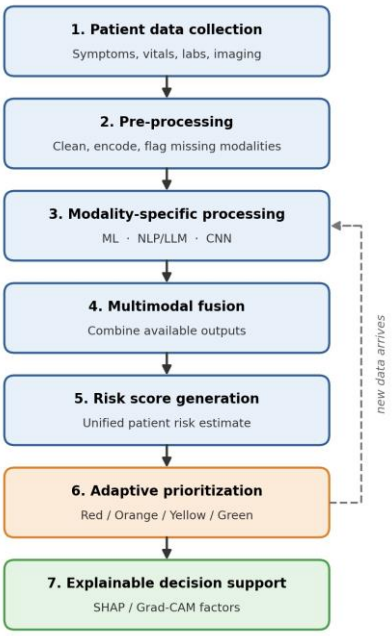


Fig. 3. Seven Step Adaptive Multimodal Triage Workflow. The dashed loop points to re-entry into the modality-specific processing component when new information from the patients becomes available.

VI. COMPARATIVE ANALYSIS

Table II presents a qualitative comparison of works from the various clusters to the desired capabilities of the proposed system. Sadly, no reviewed paper meets all the five criteria: ECG- and CXR-specific studies [1], [36] are accurate within their modality but not multimodal; disease-specific fusion work [34] is multimodal but not adaptive or missing-modality robust; studies that are more robust [45], [46] say nothing about explainability or adaptivity; and studies that are more explained, like [55] and [57] have nothing to say about fusion or adaptivity. Other ethics and governance papers [26], [28] demand a test of fairness, and do not explicitly suggest triage architectures. This trend, observed in all five clusters, is the foundation of the experience-based evidence base for the joint necessity of the four design requirements for multimodal fusion, for adaptive updating, for explainability and for the robustness in the presence of missing modalities, as well as for fairness.

TABLE II. Qualitative Comparison of Representative Prior Work

Study (ref.)	MM	Ad.	XAI	MMR	Fair
Elhussain et al. [1] (ECG-AI review)	×	×	~	×	×
Porto [11] (ML/NLP triage review)	~	×	~	×	~
Wu et al. [34] (CXR+EHR heart failure)	✓	×	×	×	×
Schouten et al. [23] (multimodal scoping review)	✓	×	~	~	~
Wang et al. [45] (missing-modality fusion)	✓	×	×	✓	×
Liaqat et al. [46] (Chameleon framework)	✓	×	×	✓	×
Pahud de Mortanges et al. [55] (XAI for multimodal/longitudinal)	~	~	✓	×	×
Hur et al. [57] (clinician-friendly XAI)	×	×	✓	×	×
Kostick-Quenet et al. [28] (multimodal AI ethics)	~	×	~	×	✓
Proposed system	✓	✓	✓	✓	✓

MM = multimodal fusion; Ad. = adaptive/dynamic updating; XAI = clinician-facing explainability; MMR = missing-modality robustness; Fair = fairness evaluation. ✓ *= addressed,* ~ *= partially addressed,* × *= not addressed.*

VII. DISCUSSION

The literature reviewed provides justification for each design objective individually but also shows that an integrated system is not easy. Research on ECG (Section II-A) and CXR (Section II-C) is now at an advanced stage and scarcely validated independently yet, to date, not been combined, whereas the surveys on multimodal-AI (Section II-B) confirms that validation of the fusion, rather than per-modality accuracy, is now the bottleneck. Although missing-modality techniques have been developed in sub-fields of the imaging world like MRI synthesis and tumor segmentation, they may not have been the most adopted techniques in the ED triage setting where the lack of modalities is not an exception but rather the rule (e.g., no ECG ordered, no imaging yet obtained). Explainability research (Section II-E) argues that clinical confidence is not only reliant on the complexity of the underlying approach to attributing the cause but more likely to be a function of how easily the result is expressed. The review of fairness has emerged from various reviews as a missing component and is therefore included in the proposed system as a mandatory evaluation step, instead of an optional add on due to the high ethic stakes as identified in Section II-B [28].

A drawback of this review is that it is limited to 59 studies were selected not for being the most important studies on ED-triage-AI—broader coverage of wearable/IoT-based continuous monitoring and federated/privacy-preserving learning was intentionally kept out of scope—but rather for their direct relevance to the four project objectives.

Additionally, evaluation practice seen in the reviewed work is inconsistent: Accuracies have been reported most frequently for ED-triage models; calibration and external validation checks have been reported for imaging models; and only a few expressly report fairness metrics for subgroups. The interpretation and evaluation of any real implementation of the proposed system should be based on results that include both per-modality performance and performance in the presence of missing data as well with results covering demographic-subgroup metrics and not a single metric of the overall accuracy.

VIII. CONCLUSION

This review organised 59 recent studies into five clusters related to AI-based multimodal diagnostic triage and connected these identified gaps with four actionable design goals: multimodal fusion, adaptive risk updating, explainable AI to the clinician, and missing modality robustness plus

fairness assessment. A system which treats "lack of information" as something to be expected, not excluded, and yields descriptions of factors, not raw model details: this evidence is all consistent. The proposed architecture will be implemented and empirically tested and fairness across demographic groups will be assessed as suggested by literature reviewed and indicated by the theory. Future extensions are wearable/IoT for continuous vital-sign monitoring, longitudinal EHR integration for richer context of clinical data for model training, voice- and multilingual-symptom input for model training accessibility and federated/multilingual model learning for secure model training across multiple healthcare institutions.

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